Birth-order sensitivity of undirforests and internodons

An exact swap theorem and finite genealogical measurements

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Abstract

The undirforest construction turns a birth-ordered genealogy into a forest on the same organisms. We identify its finite version with the augmented merge tree of the graph's superlevel components. An adjacent swap of incomparable births leaves the forest unchanged precisely when no younger component meets both organisms; otherwise it replaces exactly one clade. This gives a clade-distance bound along any sequence of valid swaps. The usual interleaving stability theorem supplies a separate bound in birth-height units. Neither bound controls the internodon membership of fixed organisms. Saved simulations illustrate the distinction: within-generation reorderings can change most parent pointers, while separating 18–31 percent of pairs that originally shared an internodon in a dedicated comparison sample. We define the denominators, finite-window conventions and sampling protocol explicitly. The results give precise partial answers to the robustness question, rather than a general biological robustness theorem.

1. One construction, several meanings of change

Alexander, de Bruin and Kornet [A] construct a forest from a genealogical directed acyclic graph with distinct birthdates. In their concluding discussion they ask whether biological undirforests depend drastically on the topological order chosen for the births. The question deliberately leaves the meaning of "drastically" open.

A forest on named organisms contains several kinds of information. It gives each organism a parent in the derived forest. It gives a hierarchy of clades. With an additional SD classification, it yields internodons. A change can be small in one of these descriptions and large in another.

The simplest example has two parents u, v and their child z:

$$u \longrightarrow z \longleftarrow v.$$

Both birth orders u, v, z and v, u, z are legitimate. Their undirforests are respectively the chains $u \to v \to z$ and $v \to u \to z$. Every parent pointer changes, but the clade families differ by just one replacement: $\{v, z\}$ becomes $\{u, z\}$. The full three-organism clade and the singleton $\{z\}$ remain.

We first prove a general version of this local calculation. We then distinguish its clade bound from an imported stability bound for height-decorated merge trees. Finally we explain the internodon measurements using a statistic that counts losses among originally co-classified pairs. That statistic and the all-pairs disagreement rate answer different questions.

The structural theorems concern finite birth-ordered graphs. The experiments are finite-window approximations to the infinite biological setting of [A]. We keep those two scopes explicit.

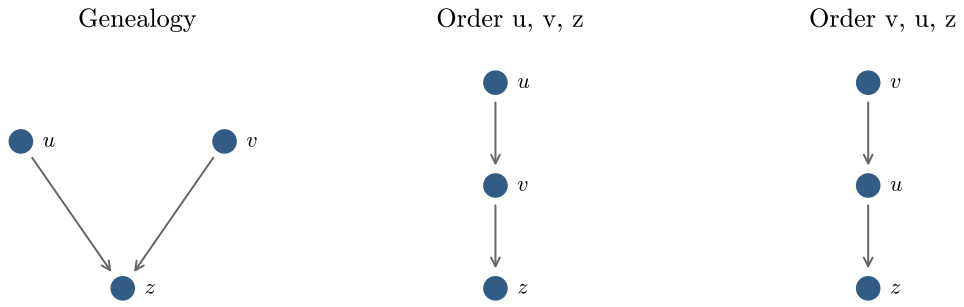


Figure 1: The same genealogy under two valid birth orders. Left: both parents precede their child. The two derived chains replace just their inner clade.

2. The undirforest as a component sweep

Let $G = (V, E)$ be a finite genealogy and let $r : V \to \{0, \ldots, n-1\}$ be a topological ranking: parents have smaller ranks than children. Write H for the underlying undirected graph.

For distinct x, y, the source defines x to be an **undirancestor** of y when an undirected path from x to y avoids every organism older than x [A, Definition 4]. Equivalently,

$$y \text{ is in the component of } x \text{ in } H[\{v : r(v) \geq r(x)\}].$$

An **undirparent** is an immediate undirancestor. By the source's Corollary 6 it is the youngest undirancestor when one exists. A vertex with no undirancestor is a root of the derived forest.

For a forest F, let $D_F[x]$ be the set containing x and all its descendants in F. Its clade family is

$$\mathcal{K}(F) = \{D_F[x] : x \in V\}.$$

Clades here retain organism labels. Every vertex supplies a distinct clade, so this family has n members.

Theorem 1. The component-sweep description

Process vertices from youngest to oldest. On adding x, merge the current components adjacent to x. Make x the parent of the oldest vertex of each merged component. The resulting forest is exactly the source's undirforest.

Proof. Immediately before adding x, the processed graph is the graph induced by vertices younger than x. Each of its components is represented in the constructed forest by its oldest vertex; all its other vertices already lie below that representative. Adding x joins precisely the components touching x.

Let y be the oldest vertex in one such component. The new connection makes x an undirancestor of y. If a younger undirancestor x', older than y, existed, its witnessing path would already lie in the processed graph. Then x' would belong to the component containing y, contrary to y being its oldest vertex. Thus x is the youngest undirancestor of y and is its undirparent.

Each non-root representative receives its parent on its first connection to an older vertex. The resulting relation therefore agrees with the source relation at every vertex. The merged component has oldest representative x, preserving the invariant for the next step. □

This is the standard component-merging sweep, retaining a vertex for every organism rather than suppressing vertices of degree two. In this augmented sense the undirforest is a merge tree on each connected component. The identification is useful because it makes the construction's dependence on the ranking visible: change the ranking and the same graph is swept in a different order.

3. Exactly what one adjacent swap changes

Let u, v be consecutive in the ranking, with u older. Assume they have no parent-child edge. Consecutiveness then makes them incomparable in the genealogical order, so their ranks can be swapped.

Let Y be the vertices younger than both. Partition the components of $H[Y]$ that touch the pair into three groups: those meeting both u, v, those meeting u only, and those meeting v only. Write S, A, B for the respective unions of their vertex sets.

Theorem 2. Adjacent-swap locality

If S is empty, the undirforest is unchanged. If S is nonempty, the clade family undergoes exactly one replacement:

$$\{v\} \cup S \cup B \quad \longmapsto \quad \{u\} \cup S \cup A.$$

The outer clade $\{u, v\} \cup S \cup A \cup B$ and every other clade stay unchanged.

More precisely, the oldest vertex of each shared component changes parent from v to u. If w was the original parent of u, then v changes parent from u to w and u changes parent from w to v. If w is absent, the corresponding vertex is a root.

Proof. In the original reverse sweep, v is processed just before u. If no component meets both, their merges are disjoint. They are not directly adjacent either. The component partition and its oldest representatives are the same after the pair in both orders; all parent assignments are the same.

If a shared component exists, adding v first makes it parent of the representatives of the shared components and the v-only components. Adding u makes it parent of the u-only representatives and of v. Its clade is the outer set, and v's clade is $\{v\} \cup S \cup B$.

In the swapped order, u is processed first. It becomes parent of the shared and u-only representatives. Then v becomes parent of u and the v-only representatives. The outer set is unchanged, and the inner clade is $\{u\} \cup S \cup A$.

After processing the pair, the same vertex set forms one component; its oldest representative is u in the first order and v in the second. Its eventual older parent w is therefore the same. All younger internal clades are unchanged, and every later merge attaches the same whole component. This proves the parent description and excludes any other clade change. The two inner clades differ because one contains v but not u, and the other contains u but not v. □

The theorem bounds the **number** of changed clades, not their sizes. Appending a long descendant chain to the shared child in the introductory example makes both replaced clades large. Using many separate shared components makes many individual parent pointers change. Neither observation contradicts the single-clade statement.

Corollary 3. A bound between any two topological orders

Let K be the number of organism pairs ordered differently by two topological rankings. Their undirforest clade families have symmetric difference at most $2K$. More precisely, along any valid adjacent-swap route with m swaps having a shared younger component, the bound is $2m$.

Proof. Transform one order into the other by successively moving the next required vertex leftwards. Each crossed pair is reversed in the two orders and is therefore incomparable in the genealogical order. This gives a valid route using exactly K swaps. Theorem 2 changes zero clades at a swap without a shared component and removes one and adds one otherwise. The triangle inequality for symmetric difference gives both bounds. □

The sharper count depends on the route's intermediate graphs. It is an exact combinatorial statement, without an assumption about a probability distribution on birth orders.

4. A separate stability statement in height units

For a connected H, extend $-r$ linearly over its edges. Its sublevel components encode the superlevel component sweep above; partial edges attach to an existing endpoint and do not create additional connections. Retain the standard infinite root ray in the associated continuous merge tree.

Morozov, Beketayev and Weber [MBW, Theorem 2] prove stability: for two scalar functions on the same domain, their merge trees satisfy $d_I(T_f, T_g) \leq \|f - g\|_\infty$. Applying it to two rankings on this same graph gives

$$d_I(T_{-r}, T_{-r'}) \leq \max_{v \in V} |r(v) - r'(v)|.$$

The edgewise difference is bounded by the maximum at endpoints. For disconnected graphs apply the statement componentwise.

This compares height-decorated trees, not fixed organism labels. The interleaving maps may move points along the trees. They need not preserve an organism's parent or internodon. A valid adjacent swap therefore has height distance at most one while still changing many parent pointers. Rank displacement is measured in birth positions, not elapsed biological time.

5. From the forest to internodons

The source's additional construction uses the **sexual descendant property**, abbreviated SD: an organism is SD if it has a strict genealogical descendant with at least two parents [A, Definition 7]. This property uses the original directed genealogy, not the derived forest.

Restrict the undirforest to SD vertices. Its maximal non-branching segments are the pre-internodons. A branching vertex ends its own segment; each of its children begins a new one. Equivalently, an SD child inherits its parent's segment exactly when that parent has only one SD child. Each non-SD organism with an SD ancestor joins the segment of its youngest SD ancestor, its test organism. These are the source's Definitions 8–9.

Non-SD organisms without an SD ancestor have no internodon under that definition. The experiment code assigns each such organism a distinct singleton label to obtain a partition of all observed vertices. These singleton labels are a reporting convention. They never create a positive co-member pair. The set of uncovered vertices is fixed across rankings of the same finite genealogy. Removing

it therefore leaves co-member loss and positive-pair Jaccard unchanged, although it changes the denominator of the all-pairs rate.

A finite window can miss later sexual descendants and later paths connecting older vertices. The source's infinite-future internodons consequently need not equal the restriction of a finite-window computation. Additional simulated generations help examine this effect, but cannot certify the infinite future.

6. What the measured percentages mean

Fix one finite genealogy and compare a reference ranking with an alternative. Let P_0 and P_1 be the sets of unordered pairs sharing an internodon label under the two rankings. We use three distinct quantities:

$$\text{all-pairs disagreement} = \frac{|P_0 \triangle P_1|}{\binom{n}{2}}, \qquad \text{reference co-member loss} = \frac{|P_0 \setminus P_1|}{|P_0|},$$

$$\text{positive-pair Jaccard} = \frac{|P_0 \cap P_1|}{|P_0 \cup P_1|}.$$

The first asks how many of all possible pair decisions change. The second asks how often a previous positive classification is lost. The third compares the two positive-pair sets symmetrically. The first two are disagreement rates; a larger Jaccard value means greater agreement. None is a percentage of individual organisms retaining a species identity.

An all-pairs rate can be small when most pairs are separated under both rankings. For example, among 100 organisms, suppose P_0 has 100 pairs, 30 are lost, and 20 new pairs appear. Only 50/4950, about one percent, of all pair decisions change, but thirty percent of the original co-memberships are lost. This is why both denominators matter.

The within-generation comparison sample

The saved co-membership experiment uses four genealogies and six alternative rankings per genealogy in each configuration, with random seed 7. There are 14 generations. One ancestral deme has N organisms per generation; from generation 6 there are two permanently isolated demes of size N each. Each child has two distinct parents with probability p and otherwise one. Parent choices are from the preceding generation in the same deme, or from the ancestral deme at the split. Alternative rankings independently permute births within each generation.

The following are means of the 24 comparisons in each row, not pooled-pair estimates or confidence intervals. They come directly from the retained co-membership record.

N	p	All-pairs disagreement	Reference co-member loss	Jaccard
6	0.3	6.70%	22.41%	0.6009
6	1.0	8.71%	18.06%	0.6750
12	1.0	6.99%	22.16%	0.6405
24	1.0	7.36%	30.83%	0.5171

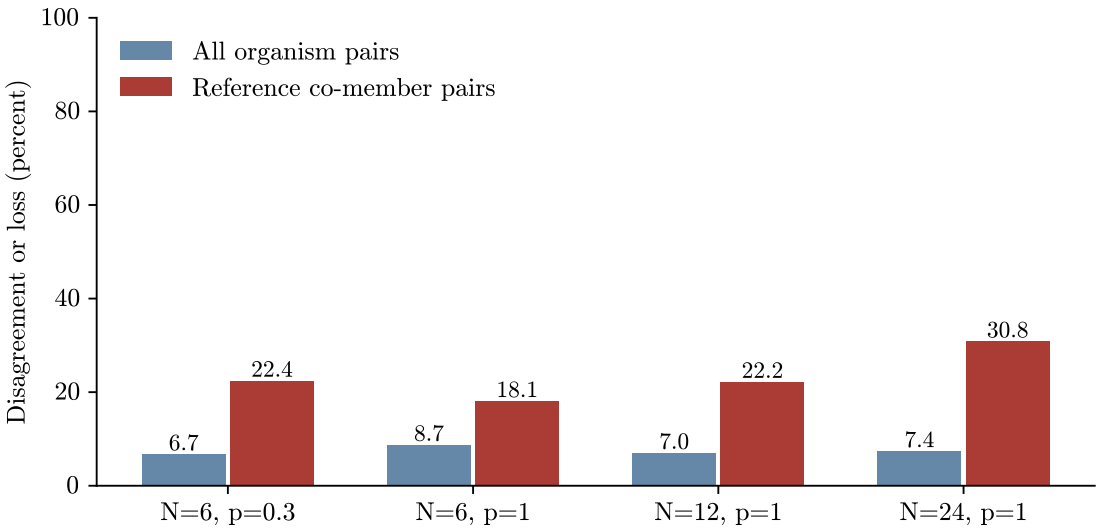


Figure 2: Both disagreement rates use the same saved comparisons. The larger rate conditions on a reference co-membership; the smaller rate averages over all organism pairs.

The largest co-member loss in an individual comparison was 60.67 percent. These measurements show appreciable reclassification in this finite model. They do not support a general claim that internodon membership is nearly unchanged.

Parent pointers and clades

A separate retained experiment uses six genealogies and twelve alternative rankings per genealogy. Its random seed is 15. The generation and split parameters are the same, but the draws differ from those in the preceding table.

N	p	Parent pointers changed	Normalized clade difference	All-pairs disagreement
6	0.3	51.00%	32.26%	7.96%
6	1.0	77.54%	58.19%	6.27%
12	0.3	56.43%	44.06%	6.07%
12	1.0	82.63%	70.30%	6.16%
24	0.3	58.69%	48.76%	4.44%
24	1.0	84.62%	73.86%	6.00%

The full record also contains $p = 0.7$. Normalized clade difference is

$$\frac{|\mathcal{K}(F_0) \triangle \mathcal{K}(F_1)|}{|\mathcal{K}(F_0)| + |\mathcal{K}(F_1)|}.$$

The parent-pointer column compares organism-labelled parents, including the root marker. The three columns measure different features. The small all-pairs numbers in this table cannot be used as substitutes for the co-member-loss numbers in the other sample.

7. Window length and the distribution on orders

Additional future generations

A buffered experiment evaluates only organisms from the first 10 generations and adds 0, 4, 8 or 12 later generations before constructing the forest and labels. Its split is at generation 4, its seed is 16, and each configuration has five genealogies and eight alternative rankings per genealogy.

For $N = 6, p = 1$, the mean number of labels on the evaluated vertices falls from 11.0 with no buffer to 5.0 with four extra generations, and is 5.4 with twelve. For $N = 12, p = 1$, the corresponding values are $20.6, 7.6, 11.0$. The label counts include the singleton convention of Section 5.

This is evidence that a short observation window affects classification. It is not a monotone convergence result: different configurations use different random draws. The buffered record contains all-pairs disagreement but no reference co-member-loss statistic. No conclusion about that conditional loss is inferred from it.

Orders spanning generations

Another retained sample chooses each next vertex uniformly from the currently available vertices in Kahn's topological sorting algorithm. This gives valid topological orders. It does **not** sample linear extensions uniformly.

For a graph with $a \to c$ and an isolated b, the possible orders abc, acb, bac have probabilities $1/4, 1/4, 1/2$. The first choice is uniform between a, b, but after choosing b the rest is forced. Thus even this small graph produces unequal extension probabilities.

With five genealogies and ten alternative orders per configuration, seed 21, the saved all-pairs disagreement ranges from 10.39 to 14.47 percent across the four configurations. Parent-pointer changes range from 56.62 to 86.73 percent. These are measurements under this particular sampling rule. They are not estimates under a uniform distribution on all admissible birth orders or under a biologically justified distribution of birthdate uncertainty.

8. What is settled, and what remains

The component sweep identifies the exact finite construction. The swap theorem gives a sharp local clade statement, and Corollary 3 gives a bound for an arbitrary pair of topological orders. The established merge-tree stability theorem gives a different bound in height units.

These are precise partial answers to the source question. They do not yield a population-independent bound on the internodon co-membership of fixed organisms. The finite measurements exhibit substantial changes, with the size depending on the statistic, observation window and order sampler. They neither establish a general biological robustness claim nor refute every possible formal version of that claim.

A stronger result about internodon membership would have to state its birthdate-error model, identify a metric on labelled classifications, and control the finite-window approximation to the future. Extending the construction to historical gene trees is a separate question from [A] and is not treated here.

Appendix. Computational records

The calculations and saved records are included with the paper.

- The direct source-definition and sweep implementations are in the accompanying undirforest program.
- The adjacent-swap record contains 4,302 valid swaps, of which 1,637 have a shared younger component. All cases satisfy Theorem 2's tested parent-change condition, and all 1,637 cases replace exactly one clade. The recorded mean replaced clade size is 58.03 percent of the observed population.
- The co-membership table and plot use the retained co-membership-check record, with 24 comparisons per row.
- The other tables and ranges use the separate main, buffered and Kahn-order records described above.

These finite checks support the implementations and the reported finite data. The universal statements rest on the proofs and the explicitly cited published stability theorem. No new biological data, independent human proof review, or historical priority assessment is claimed.

References

[A] S. A. Alexander, A. de Bruin and D. J. Kornet. *An alternative construction of internodons: the emergence of a multi-level tree of life.* Bulletin of Mathematical Biology (2015). DOI: 10.1007/s11538-014-0048-2. Definitions 4–9, Corollary 6, and Section 8.

[MBW] Dmitriy Morozov, Kenes Beketayev and Gunther H. Weber. *Interleaving Distance between Merge Trees.* Manuscript presented at TopoInVis 2013. Theorem 2, pp. 6–7. Author manuscript.